



FINANCIAL ASSISTANT APP

Business Analysis Project Documentation

Prepared by: Nino Zavrashvili

Business Analyst | Smart Academy

April 2026

TABLE OF CONTENTS

1. Executive Summary.....	p. 3
2. Business Model Canvas.....	p. 3
3. Fishbone Analysis – Pain Points & Root Causes.....	p. 7
4. User Personas.....	p.9
5. Stakeholders & RACI Matrix.....	p. 11
6. Communication Plan, CHANNELS & TOOLS.....	p. 14
7. Business Requirements & Business Rules.....	p. 18
8. User Stories (Improved with Edge Cases).....	p. 21
9. Process Diagrams (AS-IS / TO-BE).....	p. 34
10. Use Case Diagram.....	p. 36
11. User Story Map.....	p. 37
Appendix: Key Metrics & Success Criteria.....	p. 42

1. EXECUTIVE SUMMARY

Georgian population struggles with effective personal finance management. Key problems:

- Financial data scattered across multiple banking apps
- Manual expense categorization consuming significant time
- Budget control happens only at month-end (after overspending occurs)

Our solution consolidates multiple bank accounts into a single platform. The system integrates with Georgian banks via Open Banking API, uses AI to auto-categorize expenses with 85% accuracy, and provides real-time budget alerts.

2. BUSINESS MODEL CANVAS

Component	Details
Key Partners	- TBC Bank - BOG Bank - Liberty Bank - App Stores (iOS/Android) - AWS Cloud Provider - Open Banking API providers - AI/ML service providers - NBG (National Bank of Georgia)
Key Activities	- Centralization of multi-source financial data (banks, payments, accounts, offers) - Automatic transaction categorization (AI-powered, ≥85% accuracy) - Real-time notifications & AI-based recommendations - Secure & standardized Open Banking API integration - Data-driven AI analytics & financial insights Personalized AI-powered recommendations - Budget creation, control & monitoring - Platform technical support & development
Value Proposition	- All banks in one place (TBC, BOG, Liberty, etc.) - AI auto-categorization (≥85% accuracy)

	- Centralized financial management - Automatic expense categorization - Budget control & Real-time Alerts - Savings goals management - Bill Reminder automation - AI Premium advice & insights - Personalized recommendations - Periodic financial reports - Complete financial overview
Customer Relationships	- Self-service platform (24/7 access) - AI personalization & customization - AI Chatbot support - Push Alert Engagement (budget, goals) - Weekly/Monthly Financial Reports - Premium Support tier - Referral program ("Invite a friend") - In-App financial education & tips
Customer Segments	- IT/Office Workers (22-35 years) — Multiple bank accounts, investment interests - Freelancers — Irregular income, EUR/USD management, RS.ge tax payments - Families (34-40 years) — Joint budgets, loans, children expenses - Students (18-24 years) — ₾500 monthly budget, primary financial habits - Small Business Owners — Personal/business account separation
Key Resources	- iOS & Android mobile applications - Georgian Banks (TBC, BOG, Liberty, etc.) - Financial data & AI/ML models - Developer & engineering team - UI/UX Design team - Cloud Infrastructure (AWS) - API Gateway & integrations

	- NBG License / Open Banking permission
Channels	- Mobile application (iOS/Android) - Push notifications (system alerts) - Bank integrations (banners/push in bank apps) - App Store Optimization (ASO) - Search Engine Marketing (SEO/SEM) - Advertising platforms (Google Ads, Meta) - Social media (Instagram, Facebook, TikTok) - Referral program ("Invite a friend" incentives)
Cost Structure	- Personnel Costs: ₪650,000 - Platform Development: ₪140,000 - Marketing & User Acquisition: ₪25,000 - Data Security & Compliance: ₪35,000 - UX/UI Development: ₪30,000 - Legal & Regulatory Fees: ₪35,000 - Partner Integration (APIs): ₪30,000 - AI Services & ML Infrastructure: ₪40,000 - Cloud Infrastructure (AWS): ₪15,000 TOTAL ANNUAL COST: ₪1,000,000
Revenue Streams	SUBSCRIPTION MODELS: - Basic Plan: ₪15/month or ₪150/year - Premium Plan: ₪35/month or ₪350/year - Free basic features (Freemium) ADDITIONAL REVENUE: - Banking Referral commissions - Affiliate Marketing partnerships - Enterprise partnerships (B2B) Break-even point: 5,555 Premium paying users

Key partners

What are your key partners to get competitive advantage?

- TBC Bank
- BOG Bank
- Liberty Bank
- App Stores (iOS/Android)
- AWS Cloud Provider
- Open Banking API providers
- AI/ML service providers
- NBG (National Bank of Georgia)

Key activities

What are the key steps to move ahead to your customers?

- Centralization of multi-source financial data (banks, payments, accounts, offers)
- Automatic transaction categorization (AI-powered, ≥85% accuracy)
- Real-time notifications & AI-based recommendations
- Secure & standardized Open Banking API integration
- Data-driven AI analytics & financial insights
- Personalized AI-powered recommendations
- Budget creation, control & monitoring
- Platform technical support & development

Key propositions

How will you make your customers' life happier?

- All banks in one place (TBC, BOG, Liberty, etc.)
- AI auto-categorization (≥85% accuracy)
- Centralized financial management
- Automatic expense categorization
- Budget control & Real-time Alerts
- Savings goals management
- Bill Reminder automation
- AI Premium advice & Insights
- Personalized recommendations
- Periodic financial reports
- Complete financial overview

Customer relationships

How often will you interact with your customers?

- Self-service platform (24/7 access)
- AI personalization & customization
- AI Chatbot support
- Push Alert Engagement (budget, goals)
- Weekly/Monthly Financial Reports
- Premium Support tier
- Referral program ("Invite a friend")
- In-App financial education & tips

Customer segments

Who are your customers? Describe your target audience in a couple of words.

1. IT/Office Workers (22-35 years)
Multiple bank accounts, investment interests
2. Freelancers Irregular income, EUR/USD management, RS.ge tax payments
3. Families (34-40 years) Joint budgets, loans, children expenses
4. Students (18-24 years) €500 monthly budget, primary financial habits
5. Small Business Owners
Personal/business account separation

Key resources

What resources do you need to make your idea work?

- iOS & Android mobile applications
- Georgian Banks (TBC, BOG, Liberty, etc.)
- Financial data & AI/ML models
- Developer & engineering team
- UI/UX Design team
- Cloud Infrastructure (AWS)
- API Gateway & integrations
- NBG License / Open Banking permission

Channels

How are you going to reach your customers?

- Mobile application (iOS/Android)
- Push notifications (system alerts)
- Bank integrations (banners/push in bank apps)
- App Store Optimization (ASO)
- Search Engine Marketing (SEO/SEM)
- Advertising platforms (Google Ads, Meta)
- Social media (Instagram, Facebook, TikTok)
- Referral program ("Invite a friend" incentives)

Cost Structure

How much are you planning to spend on the product development and marketing for a certain period?

Personnel Costs	€650,000
Platform Development	€140,000
Marketing & User Acquisition	€25,000
Data Security & Compliance	€35,000
UX/UI Development	€30,000
Legal & Regulatory Fees	€35,000
Partner Integration (APIs)	€30,000
AI Services & ML Infrastructure	€40,000
Cloud Infrastructure (AWS)	€15,000
TOTAL ANNUAL COST:	€1,000,000

Revenue Streams

How much are you planning to earn in a certain period? Compare your costs and revenues.

SUBSCRIPTION MODELS:

- Basic Plan: €15/month or €150/year
- Premium Plan: €35/month or €350/year
- Free basic features (Freemium)

ADDITIONAL REVENUE:

- Banking Referral commissions
- Affiliate Marketing partnerships
- Enterprise partnerships (B2B)

FINANCIAL PROJECTIONS:

Break-even point: 5,555 Premium paying users

3. FISHBONE ANALYSIS – PAIN POINTS & ROOT CAUSES

Main Issue: Users Cannot Manage Personal Finances Effectively

PEOPLE FACTOR:

- Manual data entry is inconvenient
- No tracking habits established
- Lack of progress/results visibility
- No incentive mechanisms
- Does not track expenses regularly
- Inadequate financial education
- Low motivation
- Impulse spending behavior
- No systematic expense tracking

PROCESS FACTOR:

- Automation need not identified
- Lack of unified process automation
- Mismatch between planned and actual expenses
- Lack of unified payment control
- Unclear centralized reporting
- No structured budget planning

TECHNOLOGY FACTOR:

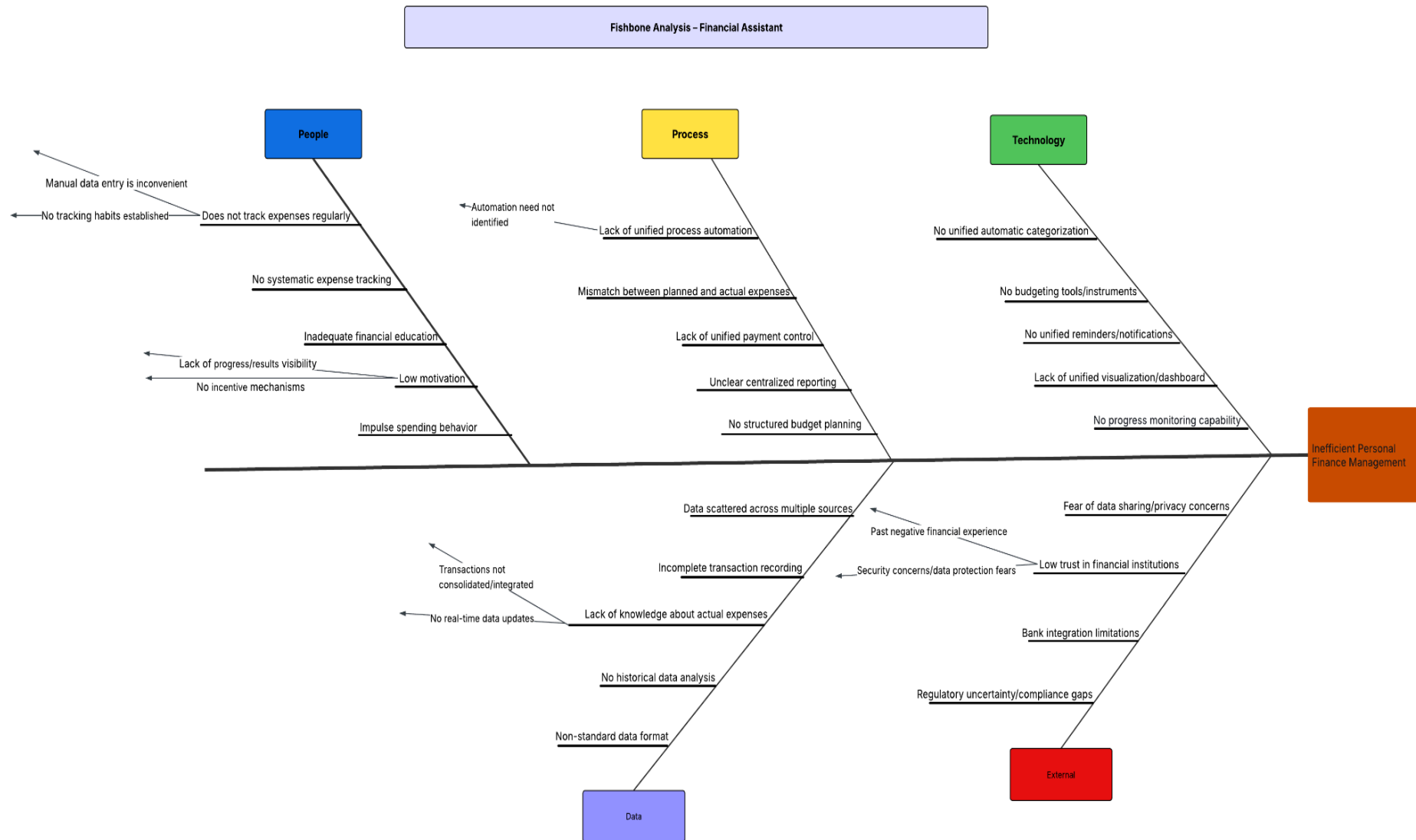
- No unified automatic categorization
- No budgeting tools/instruments
- No unified reminders/notifications
- Lack of unified visualization/dashboard
- No progress monitoring capability

DATA FACTOR:

- Data scattered across multiple sources
- Incomplete transaction recording
- Lack of knowledge about actual expenses
- No historical data analysis
- Non-standard data format
- Transactions not consolidated/integrated
- No real-time data updates

ENVIRONMENT/EXTERNAL FACTOR:

- Fear of data sharing/privacy concerns
- Low trust in financial institutions
- Bank integration limitations
- Regulatory uncertainty/compliance gaps
- Past negative financial experience
- Security concerns/data protection fears



4. USER PERSONAS

#	Name	Description	Behavior	Pain Points	Needs
UP-01	Nino	IT Specialist, 29 years old, Tbilisi. Tech-savvy professional with stable income ₾3,500+. Uses TBC and BOG banks.	- Primarily pays with cards - Frequent transactions from multiple banks - Tries to track expenses in Excel manually - Multi-currency transactions (GEL/USD) - Information scattered across different bank apps - Uses manual Excel spreadsheet (time-consuming) - Sometimes incorrect expense categorization (merchant issues)	- Different bank apps - information scattered - Manual Excel tracking takes time - Categories sometimes incorrect (merchant data) - Difficult to sum GEL/USD together - No unified view of finances - Time-consuming manual process - Currency conversion complexity	- All banks in one place (TBC, BOG, Liberty, etc.) - Automatic expense categorization - Budget limits + alerts - Simple savings planning - Multi-currency consolidation - Real-time dashboard view
UP-02	George	Freelancer, 31 years old, Batumi. Irregular income, works with multiple currencies (EUR/USD), has RS.ge (Georgian tax) obligations.	- Income varies from ₾0 to ₾5,000 per month - Doesn't know how much he can spend - No financial buffer/reserve - Often forgets to pay taxes - Spends more during "good income months" - No income forecasting system - No savings plan	- Unstable/ irregular income (0-₾5,000/month) - Doesn't know spending capacity - No financial buffer established - Tax deadline often forgotten - Overspends during good income months - Cannot forecast cash flow - Multi-currency management difficulty - RS.ge tax compliance uncertainty	- Reserve planning help - RS.ge tax payment reminder/deadline notification - Multi-currency expense consolidation - Cash flow analysis - Budget flexibility based on income - Tax deadline tracking

UP-03	Mariam & Irakli	Married couple, ages 32/35, Tbilisi. Two children. Joint household budget + home mortgage. Both earn separate incomes.	- Both spend separately - Plan joint monthly/annual budget together - Try to manually control shared expenses - Attempt expense optimization - Information/accounts scattered - Cannot see complete financial picture - Confusion with payments	- Both spend independently - Accounts/data scattered across banks - Money "disappears" - unclear where spending goes - No complete financial picture - Manual budget control is difficult - Payment chaos and confusion - Cannot track joint spending accurately - Mismatch between planned and actual expenses	- Unified view (both see same data) - Expense categorization by category - Payment reminders for joint bills - Joint reporting & detailed analysis - Shared budget tracking - Family spending pattern visibility - Transaction clarity (who spent what)
UP-04	Anna	Student, 21 years old, Tbilisi. IBSU student with stipend + parental support. Monthly budget: ₾500. Building initial financial habits.	- No budget planning - Frequent small expenses (cafes, transport) - Spontaneous/impulsive spending - No financial habits established - Doesn't know how to manage money - No spending control mechanism - First time managing own finances	- No financial education or habits - No budget planning - Frequent small untracked expenses - Impulsive spending behavior - No control over spending - Doesn't understand money management - Lacks financial literacy - Cannot see spending patterns	- Visual analytics (charts, graphs) - Basic financial tips/advice - Savings goals for small amounts - Simple tracking & education - Spending awareness/visibility - Financial literacy building - Goal-setting capability - Motivation through progress tracking
UP-05	Tornike	Small Business Owner, 47 years old, Batumi.	- Uses cash + card payments simultaneously	- No real-time data visibility - Difficult to separate cash vs card transactions	- Fast/real-time dashboard - Separated transaction tracking (personal vs business)

5. STAKEHOLDERS & RACI MATRIX

Stakeholders Overview

Stakeholder	Role	Interest	Power
Product Owner	Product vision, backlog management, release decisions	High	High
Business Analyst	Requirements gathering, documentation, stakeholder communication	High	Medium
Tech Lead	System architecture, technical decisions, dev team leadership	High	Medium
Dev Team	Feature development, bug fixes, code review	Medium	Low
AI/ML Engineer	ML model design, AI categorization system	Medium	High
QA Engineer	Quality control, test cases, bug reports, UAT coordination	Medium	Low
Bank APIs	Open Banking API connectivity & data delivery support	Medium	High
Legal/Compliance	GDPR, regulatory control, compliance management	Medium	Medium
UX/UI Designer	User experience, wireframes, prototypes, design system	High	Low
NBG (Regulator)	License & Open Banking regulation, regulatory compliance	Low	High
End Users	UAT, feedback, experience, product validation	High	Low
Director	Strategic decisions, budget approval, tech vision	High	High

RACI Matrix (Key Tasks)

Task	PO	BA	TL	DT	AI	UX	QA	BANKAPIs	NBG	LC	EU	DIR
P1.1 Product Vision	R	C	C									A
P1.2 Goals & KPIs	R	C										A
P1.3 Budget		C	C									R/A
P2.1 Req Gathering	C	R/A					I				C	
P2.2 BR Documentation	A	R	C	I						C		
P2.3 Regulatory Req		C							A	R/A		
P2.4 User Stories	A	R		I		C	C					
P2.5 Acceptance Criteria		R		C			A				C	
P3.1 System Architecture			R/A	I	C							
P3.2 API Integration Design		I	R/A	C				C				
P3.3 AI/ML Architecture					R/A							I

P3.4 UX/UI Design				I		R/A					C	
P4.1 Backend Dev			A	R			C					
P4.2 Frontend Dev			A	R		C	C					
P4.3 API Integration		C	A	R			I	C				
P4.4 AI Engine Dev					R/A							I
P4.5 Database Design			A	R						C		I
P5.1 Test Case Creation		C		C			R/A					
P5.2 Functional Testing							R					I
P5.3 Security Testing			C				R		C	A		
P5.4 Performance Testing			A	C			R					
P5.5 UAT	A	C					R				C	
P6.1 Deployment Plan	I		R	C			C			C		A
P6.2 Data Migration		C	A	R			C					I

P6.3 Monitoring Setup			A	R			C					
P6.4 User Training		R		I		C					C	A
P7.1 Communication	A	R	C				C				I	
P7.2 Issue Tracking			A	C			R				I	
P7.3 Change Management	A	C	C	I							I	R
P7.4 Compliance Monitor		C	C							R/A		I

Legend: R = Responsible / A = Accountable / C = Consulted / I = Informed

6. COMMUNICATION PLAN

#	Stakeholders	Frequency	Format/Channel	Purpose & Content	Attendees	Owner
----------	---------------------	------------------	-----------------------	------------------------------	------------------	--------------

1	Product Owner & Business Analyst	Daily (15-30 min)	Async (Slack) + Daily Standup (15 min) + Bi-weekly Sync (1 hour)	Backlog refinement, user story updates, blockers, priority shifts, requirements clarification, feedback from stakeholders	Product Owner, Business Analyst	Product Owner
2	Business Analyst & Tech Lead	Bi-weekly (1.5 hours)	Technical Meeting (Zoom/In-person)	Feasibility review of requirements, API specifications, technical constraints, architecture decisions, technical debt discussion	Business Analyst, Tech Lead, Dev Team Lead	Tech Lead
3	Business Analyst & UX/UI Designer	Bi-weekly (1 hour)	Design Review Meeting (Figma, InVision, or shared screen)	Design validation, wireframe review, user flow alignment, usability considerations, accessibility compliance, design system updates	Business Analyst, UX/UI Designer, Product Owner (optional)	UX/UI Designer
4	Business Analyst & QA Engineer	Bi-weekly (1 hour)	Test Plan Review Meeting (Jira, TestRail, or shared document)	Acceptance criteria validation, test case review, bug severity classification, UAT planning, quality standards, test coverage	Business Analyst, QA Engineer, Product Owner	QA Engineer

5	Business Analyst & End Users / User Representatives	Bi-weekly / Monthly (1-1.5 hours)	Face-to-Face (or Video Call) User Testing / Focus Groups	Requirements validation, user feedback gathering, usability testing, pain point identification, feature priority feedback, change requests	Business Analyst, End Users (5-8 representatives), Product Owner, UX/UI Designer	Business Analyst
6	Product Owner & Bank API Partners	Weekly (during integration phase) / As-needed (maintenance)	Technical Sync Call (Zoom/Teams) + Email / Documentation sharing	API specifications discussion, SLA agreements, integration support, data format alignment, sandbox access, security protocols, error handling	Product Owner, Tech Lead, Dev Team (API specialist), Bank API representatives	Tech Lead
7	Product Owner & Legal / Compliance & NBG (Regulator)	Monthly (initial phase) / As-needed (ongoing)	Regulatory Compliance Meeting (Zoom/In-person) + Email	GDPR compliance review, NBG regulatory requirements, data protection measures, compliance audit, license requirements, regulatory changes	Product Owner, Legal/Compliance Officer, Tech Lead, NBG representatives (if required)	Legal/Compliance
8	Tech Lead & All Development Team	Daily (15 min)	Daily Standup (Slack/In-person)	Progress updates, blockers identification, code review status, deployment readiness, technical issues, team alignment	Tech Lead, Dev Team, QA Engineer	Tech Lead

9	AI/ML Engineer & Dev Team	Bi-weekly (1 hour)	Technical Discussion Meeting (Zoom/In-person)	AI model performance metrics, categorization accuracy review, training data updates, model optimization, integration status, testing results	AI/ML Engineer, Dev Team (Backend), QA Engineer, Product Owner	AI/ML Engineer
10	Product Owner & Director	Weekly (1 hour) / Bi-weekly Executive Update	Executive Meeting (Zoom/In-person) + Written Status Report	Project status overview, budget tracking, resource allocation, strategic decisions, risk escalation, milestone updates, stakeholder concerns	Product Owner, Director, Tech Lead (as needed)	Product Owner
11	All Project Team (Entire Stakeholder Group)	Bi-weekly (1.5 hours)	Project Status Meeting (Zoom/Large Conference Room)	Sprint review, overall project progress, milestone updates, cross-team dependencies, announcements, Q&A, feedback gathering	Product Owner, Business Analyst, Tech Lead, Dev Team, QA Engineer, UX/UI Designer, AI/ML Engineer, Legal/Compliance, End User representatives	Product Owner
12	Business Analyst & Stakeholders (General)	Weekly (async)	Stakeholder Update Email / Slack Channel	Weekly status summary, key decisions, blockers, upcoming activities, requirements updates, calendar of events, quick announcements	All Stakeholders (email/Slack)	Business Analyst

COMMUNICATION CHANNELS & TOOLS

Channel / Tool	Description & Purpose
Slack	Real-time communication, daily team updates, rapid decision-making
Email	Official project documentation, audit records, regulatory compliance updates
Jira	Defect management, sprint task tracking, business requirements logging
Zoom/Teams	Remote video conferences, daily standups, sprint reviews
Confluence	Centralized knowledge base, technical architecture, decision logs
Google Drive	Shared project files, analytical reports, stakeholder presentations
Miro	Interactive wireframing, process modeling, team brainstorming
GitHub	Source code management, pull request reviews, technical specs

7. BUSINESS REQUIREMENTS & BUSINESS RULES

FUNCTIONAL REQUIREMENTS (FRs)

FR ID	Requirement Name	Description & Rationale	Priority
FR-01	Multi-bank Account Integration	Connect and manage accounts from multiple Georgian banks (TBC, BOG, Liberty) in a single platform.	Critical
FR-02	Flexible Pricing Model	Support free plan (1 bank limit) and premium plan (unlimited banks, advanced features).	High
FR-03	AI-Powered Financial Insights	Generate AI-powered insights and recommendations weekly to help users optimize spending.	High
FR-04	Banking Product Comparison	Compare banking products (deposits, loans, credit cards) across institutions with daily updates.	High

NON-FUNCTIONAL REQUIREMENTS (NFRs)

NFR ID	Requirement Name	Description & Rationale	Priority
NFR-01	Cross-Platform Accessibility	Accessible on both iOS and Android mobile devices.	Critical
NFR-02	Data Security & Privacy	Data encrypted in transit/rest, GDPR compliant, user data deletion on-demand.	Critical
NFR-03	Transaction Sync Performance	Data must sync automatically from connected banks within 5 minutes or less.	High
NFR-04	Regulatory Compliance	Comply strictly with National Bank of Georgia (NBG) Open Banking directives.	Critical

Business Rules

BRU ID	Rule Name	Business Rule	System Logic
BRU-01	Free Plan Limitation	Users on Free Plan can connect ONLY 1 bank account	System enforces maximum 1 connected bank for Free Plan users
BRU-02	Premium Feature Access	Premium features available ONLY to Premium Plan users	Budgets, Savings Goals, AI Insights, Alerts are restricted to Premium Plan only
BRU-03	Budget Limit Validation	Budget amount must be greater than 0	System validation: Amount > 0. Zero or negative values are rejected
BRU-04	Transaction Categorization	AI categorization accuracy target: ≥85%	If AI confidence ≥85%, category auto-applied. If <85%, present override option
BRU-05	Data Sync Schedule	Automatic sync from banks occurs max every 5 minutes	Background sync task runs every 5 minutes. Manual refresh available on-demand. Retry logic on failure
BRU-06	Budget Alert Thresholds	Alerts triggered at specific budget percentage milestones	3 threshold levels: 50%, 80%, 100% of budget spent

BRU-07	Data Retention & Deletion	Data retained for 7 years; user-requested deletion within 30 days	Regulatory compliance: 7-year retention for NBG/tax requirements. GDPR deletion: max 30 days
BRU-08	Multi-Currency Handling	All foreign currency transactions auto-converted to GEL	USD, EUR, etc. converted using current NBG exchange rate. Rate updated daily
BRU-09	Session Management	User session expires after 30 minutes of inactivity	Inactivity timer starts on app launch. Any user action resets timer. After 30 min idle → session expires

Summary of Business Rules

Plan Management: BRU-01, BRU-02 - Free vs Premium tier differentiation

Data Validation: BRU-03 - Budget amount validation; BRU-04 - AI categorization

System Operations: BRU-05 - Data sync frequency; BRU-06 - Budget alerts; BRU-09 - Session timeout

Data Governance: BRU-07 - Retention & deletion; BRU-08 - Currency conversion

Key Implementation Notes

- All Business Rules must be enforced at both application and API level
- Error messages must be user-friendly and actionable (e.g., "Upgrade to Premium" not "Premium feature required")
- Business Rules should be tested in QA as explicit test cases (one test case per rule)
- Rules BRU-07 and BRU-08 have regulatory implications (NBG, GDPR) - coordinate with Legal/Compliance
- Rules BRU-01, BRU-02 drive monetization - monitor upgrade rates and feature unlock patterns
- **Rules BRU-05, BRU-09 impact user experience - balance security/performance with usability**

8. USER STORIES

US-01 — Register Account	
<i>As a new user</i> <i>I want create an account with email verification and secure password</i> <i>So that I can securely access my financial dashboard and manage my accounts</i>	
Scenario 1: Valid registration	Given user is on registration page And has valid email and phone When user enters email "mail@sample.com" And enters phone "+995 571111111" And clicks "Send Verification Code" Then verification code sent to email And SMS sent to phone When user enters verification code And sets password "SecurePass123" Then account created successfully And user logged in automatically And dashboard displayed
Scenario 2: Invalid email	Given user is on registration page When user enters email "invalid-email" And clicks "Send Verification Code" Then error message shown: "Enter valid email (example: user@gmail.com)" And verification code NOT sent
Scenario 3: Email already exists	Given user is on registration page When user enters email "existing@gmail.com" And clicks "Send Verification Code" Then error message shown: "Email already in use. Try login or use different email"

US-02 — Connect Bank Account (OAuth)	
<i>As a user</i> <i>I want securely connect my TBC, BOG, or Liberty bank account via OAuth</i> <i>So that my financial app can automatically sync and display all my bank data</i>	
Scenario 1: Successful connection	Given user is on "Connect Bank" screen When user selects bank "TBC Bank" And clicks "Connect" Then user redirected to TBC OAuth login When user logs in with bank credentials And grants app permission to access accounts Then redirected back to app And success message: "TBC account connected" And account appears in dashboard And data syncs within 5 minutes
Scenario 2: Cancel OAuth flow	Given user is on TBC OAuth login page When user clicks "Cancel" or browser back button Then redirected back to app And message shown: "Connection cancelled. Try again" And account NOT added
Scenario 3: Connection timeout	Given user connecting to bank When connection takes >30 seconds Then system auto-retries (attempt 1 of 3) And loading spinner shown: "Connecting... Attempt 2 of 3" If timeout after 3 attempts: Then error shown: "Connection timeout. Check internet and try again"

US-03 — View Dashboard	
<i>As a user</i> <i>I want see a unified dashboard of all my bank accounts, balances, and spending</i> <i>So that I can get a complete financial overview in real-time</i>	
Scenario 1: Dashboard with 2 banks	Given user has 2 banks connected (TBC + BOG) When user opens app Then dashboard loads in <2 seconds And displays: - Total balance ₺15,430.50 - Breakdown by bank (TBC: ₺10,000 BOG: ₺5,430.50) - Monthly spending chart (visual) - Latest transactions (last 7 days) - Budget deadlines (next ending date)
Scenario 2: No banks connected	Given user with 0 banks connected When user opens app Then empty state shown And message: "Connect a bank to get started" And prominent "Connect Bank" button displayed And tutorial/walkthrough available
Scenario 3: Data sync in progress	Given user opens dashboard And data sync happening in background When sync is in progress Then loading indicator shown And message: "Updating..." And partial data shown (cached from last sync)

US-04 — View Transaction History	
<i>As a user</i> <i>I want view, search, and filter all my transactions by date, merchant, and category</i> <i>So that I can understand my spending patterns and find specific transactions</i>	
Scenario 1: View and filter	Given user has 150+ transactions When user enters date range "Sep 1 - Sep 30" Then system shows transactions from that period And transaction list displays: Date, Merchant, Amount, Category, Status And count shown: "Showing 28 of 150 transactions"
Scenario 2: Search merchant	Given transaction history page open When user enters "Carrefour" in search Then system filters transactions And shows only Carrefour transactions And count: "5 transactions from Carrefour"
Scenario 3: Foreign currency	Given transaction: "€100 payment to Booking.com" When user views transaction Then displays: "€100 = ₵270.50 GEL" And shows conversion rate used (NBG rate)

US-05 — Auto-Categorize Transactions

As a user

I want have my transactions automatically categorized by AI with 85%+ accuracy

So that I can understand my spending by category without manual effort

Scenario 1: High confidence

Given new transaction: "Carrefour GEO 250.50"
When AI analyzes transaction
And AI confidence: 95%
Then category auto-applied: "Groceries"
And no user action required
And transaction appears categorized in list

Scenario 2: Low confidence

Given new transaction: "ABC Store 150.00"
When AI analyzes transaction
And AI confidence: 65% (uncertain)
Then category shown as "Uncertain"
And user shown options: [Groceries, Shopping, Other]
When user selects correct category
Then category applied
And AI learns from correction

Scenario 3: Override incorrect

Given transaction wrongly categorized as "Groceries"
When user taps category
Then category options shown
When user selects "Transport"
Then category changed immediately
And system logs correction for AI training

US-06 — Create Budget	
<i>As a user</i> <i>I want create monthly budgets by spending category with validation</i> <i>So that I can control my spending and stay within financial limits</i>	
Scenario 1: Valid budget	Given user on "Create Budget" screen When user selects category "Food" And enters amount "2000" And selects period "Monthly" And clicks "Create" Then budget saved successfully And message shown: "Food budget created" And budget appears in dashboard And budget card shows: "Food: ₪2,000/month"
Scenario 2: Negative amount	Given user on budget creation screen When user enters amount "-50" Then error shown: "Budget must be positive (e.g., 500 ₪)" And save button DISABLED And cursor in amount field (for correction)
Scenario 3: Mid-month budget	Given today is Sep 15, budget period "Monthly" When user creates "Food: ₪2,000" budget Then system calculates pro-rata: ₪1,032 for remaining Sep And message: "For Sep 15-30: ₪1,032 (pro-rata). Full ₪2,000 starting Oct 1"

US-07 — Track Budget Progress	
<i>As a user</i> <i>I want see real-time progress of my spending against budget limits</i> <i>So that I know when I am approaching or exceeding my budget</i>	
Scenario 1: Progress at 60%	Given Food budget: ₺2,000/month And user spent: ₺1,200 When user views budget progress Then displays: Spent ₺1,200, Remaining ₺800, Percentage 60% And progress bar: 60% filled (green color) And days remaining: "8 days left in Sep"
Scenario 2: Progress at 80% (warning)	Given budget progress: 80% When user views budget Then progress bar colored YELLOW (warning) And message shown: "WARNING: You've spent 80% of budget"
Scenario 3: Budget exceeded	Given budget: ₺2,000 And spending: ₺2,500 When user views budget Then progress bar colored RED And displays: "EXCEEDED by ₺500" And message: "Over budget. Plan to adjust or upgrade plan"

US-08 — Receive Budget Alerts	
<i>As a user</i> <i>I want receive push notifications when I reach budget thresholds (50%, 80%, 100%)</i> <i>So that I am alerted in real-time about my spending status</i>	
Scenario 1: Alert at 50%	Given Food budget: ₪2,000 And spending reaches: ₪1,000 When system detects 50% threshold Then push notification sent: "Halfway through Food budget" And notification content: "₪1,000 of ₪2,000 spent (50%)" And notification appears in notification center
Scenario 2: Alert at 80%	Given Food spending now: ₪1,600 When 80% threshold reached Then push notification: "WARNING: 80% of Food budget spent" And content: "₪1,600 of ₪2,000 remaining: ₪400"
Scenario 3: One alert per threshold	Given budget at 80% When ₪100 more spent (still 84%) Then NO alert sent (already alerted at 80%) And spam prevented

US-09 — Set Savings Goals	
<i>As a user</i> <i>I want create savings goals with targets and target dates</i> <i>So that I can plan and track progress toward saving for specific objectives</i>	
Scenario 1: Create goal	Given user on "Savings Goals" screen When user creates goal: - Goal Name: "Vacation Fund" - Target Amount: "€5,000" - Target Date: "Aug 15, 2024" And clicks "Create" Then goal saved And displays: - Saved Amount: €0 - Target: €5,000 - Progress: 0% - Suggested Monthly: €416/month - Days Remaining: 245 days
Scenario 2: Add money to goal	Given Vacation Fund goal exists When user transfers €1,000 Then goal progress updated: - Saved: €1,000 - Progress: 20% - Remaining: €4,000 - New Suggested Monthly: €327/month

Scenario 3: Goal completed early	Given goal: ₪5,000 by Aug 15 And user saved ₪5,000 on Aug 5 When goal amount reached Then celebration shown: "Goal achieved!" And message: "Vacation Fund completed 10 days early" And options: [Mark as Completed] [Continue Saving]
---	---

US-10 — Manage Plan Subscription	
<i>As a user</i> <i>I want upgrade from Free to Premium plan and manage my subscription</i> <i>So that I can unlock premium features like multiple bank accounts and advanced analytics</i>	
Scenario 1: Upgrade to Premium	Given user on Free plan And has 1 bank connected When user taps "Upgrade to Premium" Then pricing screen shown: - Free: ₪0 (current) - Premium: ₪35/month or ₪350/year When user selects "Premium ₪35/month" And enters payment info And clicks "Subscribe" Then subscription activated And message: "Welcome to Premium!" And Premium features unlocked immediately

Scenario 2: Downgrade to Free	Given user on Premium plan with 2 banks When user goes to Settings > Plan > "Downgrade" Then warning shown: "You have 2 banks. Free plan allows 1." And instruction: "Disconnect a bank before downgrading" When user disconnects 1 bank: Then "Downgrade" button enabled When user clicks "Downgrade" Then downgrade executed And Premium features disabled And message: "You've downgraded to Free"
Scenario 3: Payment fails	Given user Premium subscription active When payment fails on billing date Then email sent: "Payment failed. Update payment method" And grace period: 3 days to retry When user updates payment info: Then charge retried immediately If still fails after 3 days: Then subscription paused

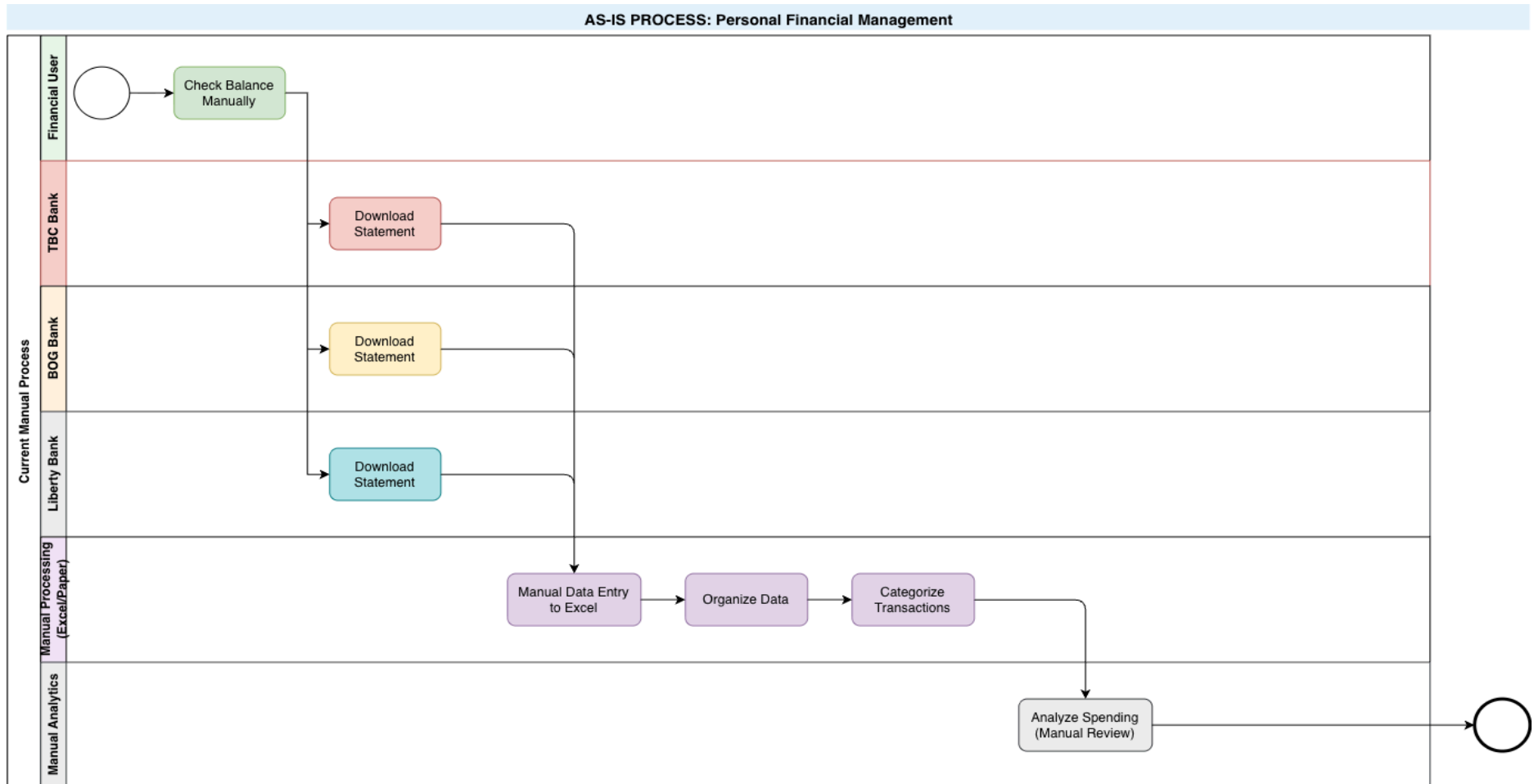
US-11 — Add Manual Expense	
<i>As a user</i> <i>I want manually add cash and receipt expenses to my account</i> <i>So that all spending is tracked and counted toward my budgets</i>	
Scenario 1: Valid expense	Given user on Dashboard When user taps "Add Expense" Then manual entry form shown with fields: - Amount - Category - Date - Merchant (optional) When user fills all and taps "Save" Then expense added immediately And appears in transaction list And counts toward budget And message: "Expense added"
Scenario 2: Zero amount	Given manual entry form open When user enters amount "0" And clicks "Save" Then error shown: "Enter amount greater than 0 (e.g., 50 ₪)" And save button DISABLED
Scenario 3: Negative amount	Given user enters amount "-50" Then error shown: "Amount cannot be negative. Did you mean refund/return?" And buttons shown: [Cancel] [Add as Refund]

US-12 — View AI Insights & Recommendations	
<i>As a user</i> <i>I want receive weekly AI-generated insights about my spending patterns and trends</i> <i>So that I can make informed financial decisions and optimize my spending</i>	
Scenario 1: Weekly insights	Given user has 2+ weeks of transaction data When user opens "Insights" tab Then weekly insights displayed: - Spending Trend: "Food spending ↑ 15% vs last week" - Category Highlight: "Top spending: Transport (€500)" - Savings Rate: "Current: 12% (was 10% last week)" - Recommendations: "Save €200/month by reducing subscriptions" And insights updated every Monday
Scenario 2: Insufficient data	Given user registered 5 days ago When user views Insights Then message shown: "Not enough data yet" And progress: "2 weeks needed for insights (5 days remaining)" And tip: "Keep spending to unlock personalized insights"
Scenario 3: Unusual spending spike	Given normal food spending: €500/week And this week: €1,200 food When AI detects anomaly Then insight: "Unusual spike in Food (€1,200 vs €500 usual)" And prompt: "Want to investigate details?"

9. PROCESS DIAGRAMS (AS-IS / TO-BE)

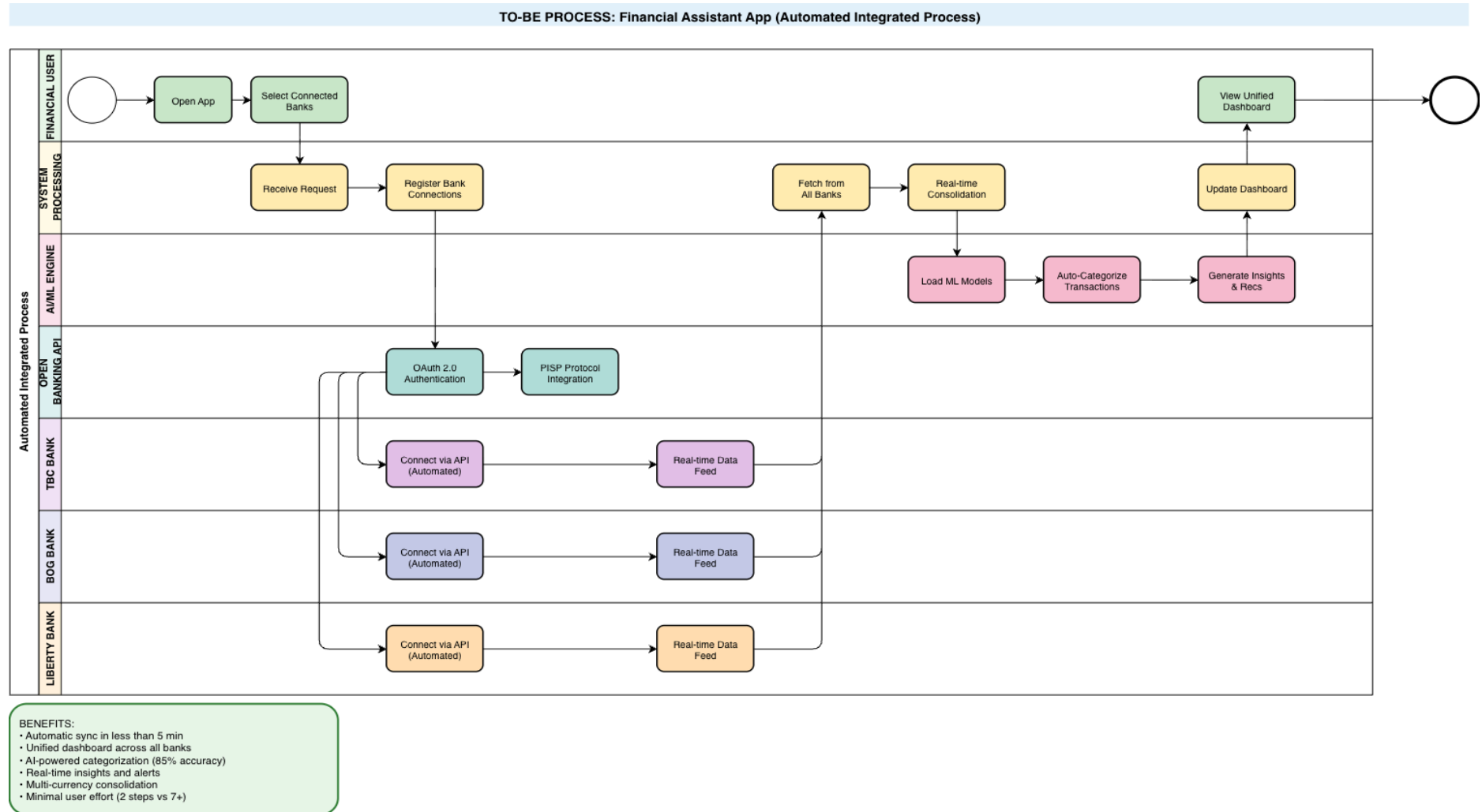
AS-IS Diagram

Current state: 3 isolated banking workflows. No consolidation, no automation, budget tracking only at month-end.

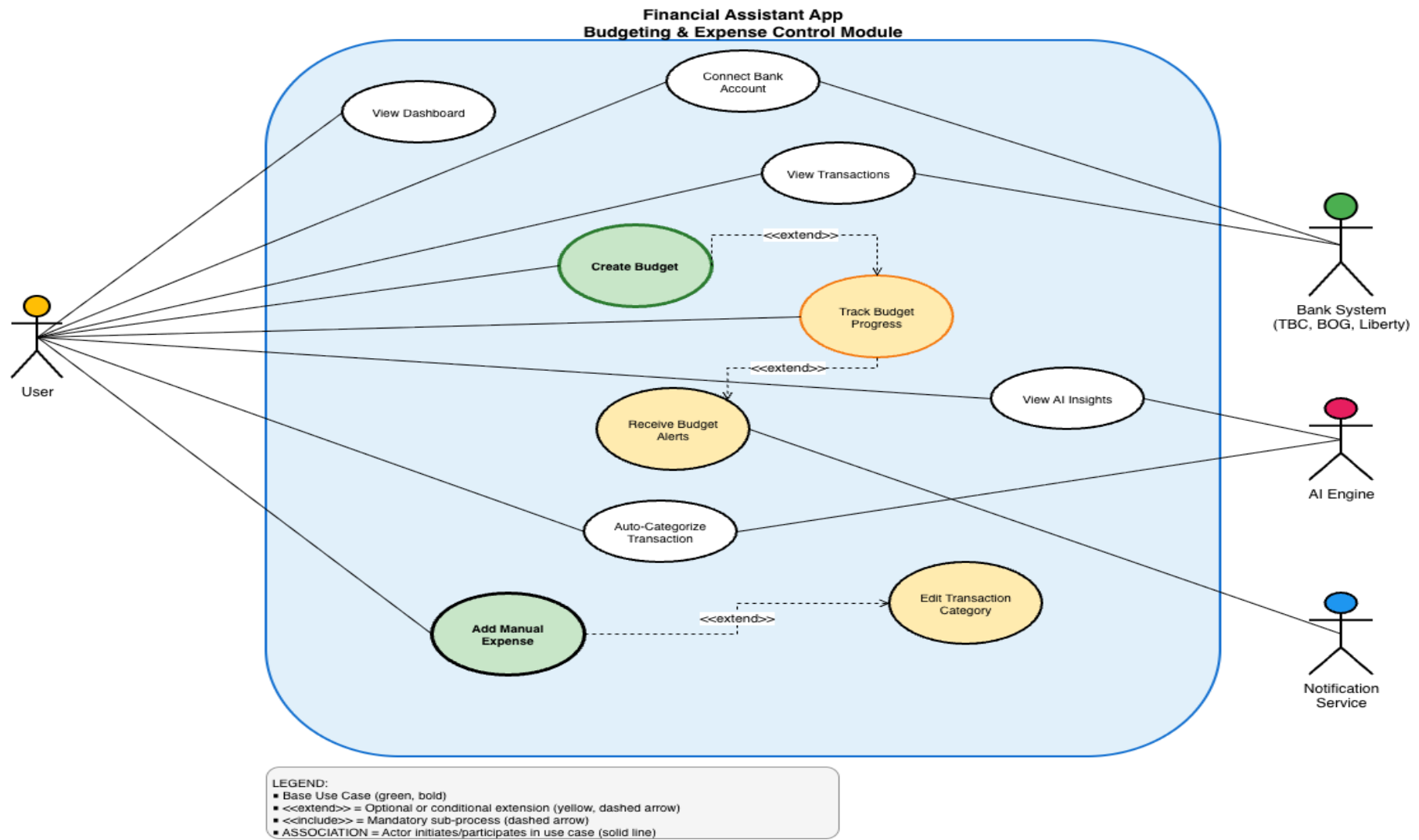


TO-BE Diagram

Improved state: Consolidated platform with automated categorization, real-time alerts, AI insights.



10. USE CASE DIAGRAM (CORRECTED)



11. USER STORY MAP - RELEASE STRUCTURE

USER STORY MAP - RELEASE STRUCTURE EPICS (5 Main Feature Areas)

Epic ID	Epic Name & Description
Epic 1	ONBOARDING User registration, login, authentication, account setup, email verification
Epic 2	BANKING Connect bank accounts via OAuth, view dashboards, sync transactions from multiple banks
Epic 3	ANALYTICS Auto-categorize transactions, AI insights, spending trends, accuracy reporting
Epic 4	BUDGET & GOALS Create budgets, track progress, set savings goals, manage subscriptions
Epic 5	NOTIFICATIONS & ALERTS Budget alerts at thresholds (50%, 80%, 100%), push notifications

RELEASE 1 (MVP) - MUST HAVE

Critical features required for MVP launch

Priority	User Story ID	Feature Name
1	US-01	Register Account & Email Verification
2	US-02	Connect Bank Account (OAuth - TBC, BOG, Liberty)
3	US-03	View Unified Dashboard
4	US-04	View Transaction History
5	US-05	Auto-Categorize Transactions (AI)
6	US-06	Create Monthly Budgets
7	US-09	Set Savings Goals
8	US-08	Budget Alerts (50% & 80% thresholds)

RELEASE 1 (MVP) - SHOULD HAVE

Important features for enhanced MVP (if time permits)

Priority	Feature ID	Feature Name
9	PWD-REC	Password Recovery/Reset
10	MULTI-ACC	Multi-Account View (dashboard with all banks)
11	US-11	Add Manual Expense Entry
12	CHARTS	Spending Charts & Visualizations
13	BUDGET-HST	Budget History & Previous Period Comparison
14	US-08-100	Budget Alert at 100% (exceeded)
15	NOTIF-CTR	Notification Center

RELEASE 1 (MVP) - NICE TO HAVE

Optional enhancements (bonus if time allows)

Priority	Feature ID	Feature Name
16	LANG-SEL	Language Selection (Georgian/English/Russian)
17	TREND-CHT	Advanced Trend Analysis Charts
18	US-12	AI Insights & Recommendations
19	GOAL-CEL	Goal Achievement Celebration
20	PDF-EXP	PDF Report Export

RELEASE 2 (Advanced Features)

Post-MVP enhancements for scaling

Feature ID	Feature Name
FAMILY-SHARE	Family Shared Account (multi-user budget tracking)
INVOICE-CREATE	Invoice & Receipt Management
TAX-AUTO	RS.ge Tax Automation (quarterly filing)
WEB-VERSION	Web Version (responsive design)

RELEASE 3 (Future Vision)

Long-term roadmap features

Feature ID	Feature Name
CRYPTO-WALLET	Cryptocurrency Wallet Integration
AI-PREDICT	Advanced AI Spending Predictions
MOBILE-PAY	In-App Mobile Payments

RELEASE SUMMARY

Release	Timeline	Feature Count	Key Objective
R1 (MVP)	Months 1-3	20 features	Launch core financial management app
R2 (Advanced)	Months 4-6	4 features	Multi-user & tax automation
R3 (Vision)	Months 7+	3 features	Crypto & AI predictions

APPENDIX: KEY METRICS & SUCCESS CRITERIA

Metric	Target Value	Measurement Tool / Method
AI Categorization Accuracy	≥85%	Database logs (tracking user manual re-categorizations vs AI predictions)
Transaction Sync Time	≤5 minutes	APM Monitoring (Datadog / AWS CloudWatch) tracking API response times
User Retention (30-day)	≥60%	Product Analytics (Mixpanel / Amplitude / GA4 cohort analysis)
Free-to-Premium Conversion	≥15%	Funnel analysis in Mixpanel & Payment Gateway (Stripe / TBC e-commerce)
App Rating	≥4.5 stars	App Store Connect & Google Play Console analytics
Average Session Duration	≥5 minutes	Product Analytics (Mixpanel / Google Analytics 4)
Budget Overrun Reduction	↓40%	Internal BI / SQL queries (comparing planned budget vs actual spent)
API Uptime	99.9%	Infrastructure monitoring tools (Datadog / Pingdom)

Document Version: 1.0 (English)

Last Updated: April 2026

Status: Approved for Development Phase

Prepared by: Nino Zavrashvili
Business Analyst | Smart Academy